

Secondary Voltage Control - MPC for the Sub Transmission Network

Sensitivity matrices computation

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Le réseau
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Net injected power at bus i

Algebraic quantity representing the local balance between generation and load:

$$P_i^{\text{inj}} = P_i^{\text{gen}} - P_i^{\text{load}}, \quad Q_i^{\text{inj}} = Q_i^{\text{gen}} - Q_i^{\text{load}}$$

It is **positive** if the bus generates (generation dominant), **negative** if it consumes (load dominant).

Load flow Jacobian

The power flow equations are nonlinear. The Jacobian JAC **linearises** them in the neighbourhood of an operating point:

$$\begin{bmatrix} \Delta P^{\text{inj}} \\ \Delta Q^{\text{inj}} \end{bmatrix} = \underbrace{\begin{bmatrix} \partial P / \partial \theta & \partial P / \partial |V| \\ \partial Q / \partial \theta & \partial Q / \partial |V| \end{bmatrix}}_{JAC} \begin{bmatrix} \Delta \theta \\ \Delta |V| \end{bmatrix}$$

Each block quantifies the sensitivity of injected powers to the state variables $(\theta, |V|)$.

Decoupling assumption:

In practice, reactive power is mainly governed by voltage magnitudes, and active power by voltage angles. The cross-sensitivity blocks are therefore small and are neglected:

$$\frac{\partial P}{\partial |V|} \approx 0, \quad \frac{\partial Q}{\partial \theta} \approx 0$$

Sensitivity Matrices: C_v, C_q

Sensitivity matrices are computed with subblocks of $\frac{\partial Q}{\partial V}$ which is contained in JAC .

C_v, C_q represent the variation of voltage in PQ buses and the reactive power in generators when **only the voltages at PV buses are varied**.

PQ buses ($\Delta Q_{pq}^{inj} = 0$):

$$0 = \Delta Q_{pq}^{inj} = \frac{\partial Q_{pq}^{inj}}{\partial V_{pq}} \Delta V_{pq} + \frac{\partial Q_{pq}^{inj}}{\partial V_{pv}} \Delta V_{pv}$$
$$\Rightarrow \boxed{C_v = \frac{\Delta V_{pq}}{\Delta V_{pv}} = - \left(\frac{\partial Q_{pq}^{inj}}{\partial V_{pq}} \right)^{-1} \frac{\partial Q_{pq}^{inj}}{\partial V_{pv}}} \quad (1)$$

PV buses ($\Delta Q_{pv}^{inj} = \Delta Q_{pv}^{gen}$):

$$\Delta Q_{pv}^{gen} = Q_{pv}^{inj} = \frac{\partial Q_{pv}^{inj}}{\partial V_{pq}} \Delta V_{pq} + \frac{\partial Q_{pv}^{inj}}{\partial V_{pv}} \Delta V_{pv} = \left(\frac{\partial Q_{pv}^{inj}}{\partial V_{pv}} + \frac{\partial Q_{pv}^{inj}}{\partial V_{pq}} C_v \right) \Delta V_{pv}$$

Hence, the second sensitivity matrix is defined as

$$\boxed{C_q = \frac{\Delta Q_{pv}^{gen}}{\Delta V_{pv}} = \frac{\partial Q_{pv}^{inj}}{\partial V_{pv}} + \frac{\partial Q_{pv}^{inj}}{\partial V_{pq}} C_v} \quad (2)$$

Nodal admittance matrix

The network topology and branch parameters are summarised in the admittance matrix

$\mathbf{Y} = \mathbf{G} + j\mathbf{B} \in \mathbb{C}^{n_{\text{bus}} \times n_{\text{bus}}}$, which relates the vector of injected currents to the vector of bus voltages:

$$\mathbf{I} = \mathbf{Y} \mathbf{V}$$

Link with the complex power injection S

The complex power injected at bus i is $S_i^{\text{inj}} = P_i^{\text{inj}} + jQ_i^{\text{inj}} = V_i I_i^*$, so that

$$S_i^{\text{inj}} = V_i \left(\sum_k Y_{ik} V_k \right)^*$$

Expression of P and Q

Writing $V_i = |V_i|e^{j\theta_i}$, $Y_{ik} = G_{ik} + jB_{ik}$ and $\theta_{ik} = \theta_i - \theta_k$, the real and imaginary parts of S_i^{inj} give:

$$P_i^{\text{inj}} = |V_i| \sum_k |V_k| (G_{ik} \cos \theta_{ik} + B_{ik} \sin \theta_{ik}), \quad Q_i^{\text{inj}} = |V_i| \sum_k |V_k| (G_{ik} \sin \theta_{ik} - B_{ik} \cos \theta_{ik})$$

Small-angle approximation ($\cos \theta_{ik} \approx 1$, $\sin \theta_{ik} \approx 0$):

$$P_i^{\text{inj}} \approx |V_i| \sum_k |V_k| G_{ik} \quad Q_i^{\text{inj}} \approx -|V_i| \sum_k |V_k| B_{ik}$$

Reactive Power Injections of an OLTC Branch

For a branch (pure reactance X , tap ratio $\tau = \frac{HV}{LV}$ in matpower), the admittance stamp in the nodal matrix is:

$$Y = \frac{1}{jX} \begin{pmatrix} \text{HV} & \text{LV} \\ \frac{1}{\tau^2} & -\frac{1}{\tau} \\ -\frac{1}{\tau} & 1 \end{pmatrix} \begin{matrix} \text{HV} \\ \text{LV} \end{matrix}$$

Reactive power injections at the HV-bus and LV-bus (small-angle approximation $\sin(\theta_{HV} - \theta_{LV}) \approx 0$):

$$Q_{HV}^{\text{inj}} = \frac{V_{HV}^2}{X \tau^2} - \frac{V_{HV} V_{LV}}{X \tau}, \quad Q_{LV}^{\text{inj}} = \frac{V_{LV}^2}{X} - \frac{V_{LV} V_{HV}}{X \tau}$$

Differentiating with respect to τ at fixed voltages:

$$\frac{\partial Q_{HV}^{\text{inj}}}{\partial \tau} = -\frac{2 V_{HV}^2}{X \tau^3} + \frac{V_{HV} V_{LV}}{X \tau^2}, \quad \frac{\partial Q_{LV}^{\text{inj}}}{\partial \tau} = \frac{V_{LV} V_{HV}}{X \tau^2}$$

Assembling over all branches defines the matrix:

$$\frac{\partial Q^{\text{inj}}}{\partial \tau} \in \mathbb{R}^{n_{\text{bus}} \times n_{\text{branch}}}$$

C_v^{tap} , C_q^{tap} represent the variation of voltage in PQ buses and the reactive power in generators when the tap is modified and are computed assuming **all power injections and voltages at PV buses remain unchanged**.

PQ buses ($\Delta Q_{pq}^{\text{inj}} = 0$):

$$0 = \Delta Q_{pq}^{\text{inj}} = \frac{\partial Q_{pq}^{\text{inj}}}{\partial V_{pq}} \Delta V_{pq} + \frac{\partial Q_{pq}^{\text{inj}}}{\partial \tau} \Delta \tau$$

$$\Rightarrow \boxed{C_v^{\text{tap}} = \frac{\Delta V_{pq}}{\Delta \tau} = - \left(\frac{\partial Q_{pq}^{\text{inj}}}{\partial V_{pq}} \right)^{-1} \frac{\partial Q_{pq}^{\text{inj}}}{\partial \tau}} \quad (3)$$

PV buses ($\Delta Q_{pv}^{\text{inj}} = \Delta Q_{pv}^{\text{gen}}$):

$$\Delta Q_{pv}^{\text{gen}} = \Delta Q_{pv}^{\text{inj}} = \frac{\partial Q_{pv}^{\text{inj}}}{\partial V_{pq}} \Delta V_{pq} + \frac{\partial Q_{pv}^{\text{inj}}}{\partial \tau} \Delta \tau = \left(\frac{\partial Q_{pv}^{\text{inj}}}{\partial V_{pq}} C_v^{\text{tap}} + \frac{\partial Q_{pv}^{\text{inj}}}{\partial \tau} \right) \Delta \tau$$

$$\Rightarrow \boxed{C_q^{\text{tap}} = \frac{\Delta Q_{pv}^{\text{gen}}}{\Delta \tau} = \frac{\partial Q_{pv}^{\text{inj}}}{\partial V_{pq}} C_v^{\text{tap}} + \frac{\partial Q_{pv}^{\text{inj}}}{\partial \tau}} \quad (4)$$

Reactive Power Injections of a Shunt Capacitor

Adding a shunt capacitor of susceptance $\Delta B_{sh,k} > 0$ at bus k is equivalent to incrementing the shunt susceptance of that bus:

$$B_{sh,k} \leftarrow B_{sh,k} + \Delta B_{sh,k}$$

The resulting reactive power injection at bus k is:

$$Q_k^{\text{inj}} = -B_{sh,k} V_k^2$$

Differentiating with respect to $B_{sh,k}$ at fixed voltages:

$$\frac{\partial Q_k^{\text{inj}}}{\partial B_{sh,k}} = -V_k^2$$

Assembling over all buses defines the matrix:

$$\frac{\partial Q^{\text{inj}}}{\partial B_{sh}} = \text{diag}(V^2) \in \mathbb{R}^{n_{\text{bus}} \times n_{\text{bus}}}$$

which is diagonal since the shunt at bus k only affects Q_k^{inj} directly.

C_v^{cap} , C_q^{cap} represent the variation of voltage in PQ buses and the reactive power in generators when a shunt susceptance is modified, assuming **all power injections and voltages at PV buses remain unchanged**.

PQ buses ($\Delta Q_{pq}^{\text{inj}} = 0$):

$$0 = \Delta Q_{pq}^{\text{inj}} = \frac{\partial Q_{pq}^{\text{inj}}}{\partial V_{pq}} \Delta V_{pq} + \frac{\partial Q_{pq}^{\text{inj}}}{\partial B_{sh}} \Delta B_{sh}$$

$$\Rightarrow \boxed{C_v^{\text{cap}} = \frac{\Delta V_{pq}}{\Delta B_{sh}} = - \left(\frac{\partial Q_{pq}^{\text{inj}}}{\partial V_{pq}} \right)^{-1} \frac{\partial Q_{pq}^{\text{inj}}}{\partial B_{sh}}} \quad (5)$$

PV buses ($\Delta Q_{pv}^{\text{inj}} = \Delta Q_{pv}^{\text{gen}}$):

$$\Delta Q_{pv}^{\text{gen}} = \Delta Q_{pv}^{\text{inj}} = \frac{\partial Q_{pv}^{\text{inj}}}{\partial V_{pq}} \Delta V_{pq} + \frac{\partial Q_{pv}^{\text{inj}}}{\partial B_{sh}} \Delta B_{sh} = \left(\frac{\partial Q_{pv}^{\text{inj}}}{\partial V_{pq}} C_v^{\text{cap}} + \frac{\partial Q_{pv}^{\text{inj}}}{\partial B_{sh}} \right) \Delta B_{sh}$$

$$\Rightarrow \boxed{C_q^{\text{cap}} = \frac{\Delta Q_{pv}^{\text{gen}}}{\Delta B_{sh}} = \frac{\partial Q_{pv}^{\text{inj}}}{\partial V_{pq}} C_v^{\text{cap}} + \frac{\partial Q_{pv}^{\text{inj}}}{\partial B_{sh}}} \quad (6)$$